

Madimetja Reginald Maredi

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Professional Profile

MSc Candidate in Mathematical Statistics at the University of the Witwatersrand. My research focuses on nonparametric estimation via Bernstein copulas, specifically using Bernstein copulas to analyse how different variables interact and influence each other. I bridge the gap between advanced statistical theory and computational implementation using R, Python, and SQL, with an interest in quantitative modelling and data science.

Technical Skills

- **Analysis & Coding:** R, SQL, Python, MATLAB, C#.
- **Actuarial & Data Science:** Stochastic Processes, Bayesian Analysis, Multivariate Analysis, Monte Carlo Simulation.
- **Tools:** Microsoft Excel, Git.
- **Key Competencies:** Statistical Inference, Data Cleaning, Algorithm Development, Object-Oriented Design.

Education

MSc in Mathematical Statistics (Dissertation only) 2026–Present
University of the Witwatersrand

- **Dissertation:** Non-parametric estimation via Bernstein copulas.
- **Relevance:** Researching dependence structures between variables, a key statistical method used in actuarial science for modelling joint risk events and portfolio aggregation.

BSc Hons in Mathematical Statistics 2025
University of the Free State (UFS)

BSc in Mathematics and Statistics 2022–2024
Nelson Mandela University (NMU)

Relevant Experience

Statistics III (STSA3716) Teaching Assistant Mar 2025–June 2025
University of the Free State

- **Data Accuracy:** Responsible for grading assignments with strict attention to detail, ensuring assessment data is recorded accurately and fairly for third-year students.
- **Communication:** Facilitate tutorials and provide written feedback to help students understand complex statistical concepts, bridging the gap between technical theory and practical application.

Mathematics for Engineers (MATV111) Teaching Assistant Feb 2024–June 2024
Nelson Mandela University

- **Leadership & Mentoring:** Supported a cohort of over 100 students, acting as the primary point of contact for academic support and troubleshooting.
- **Consultation:** Conducted one-on-one sessions to diagnose learning gaps and provide tailored solutions to improve student performance.

- **Analysis:** Evaluated student output to identify common error patterns and provided feedback to the lecturer to adjust teaching strategies.

Volunteering & Community Impact

- **Community Tutor** (Kgagatlou, Thagaetala, Dikobe Molaba Schools): Dedicated time to tutoring high school Mathematics and Physical Sciences, helping learners improve their grasp of core STEM subjects.
- **Community Mentor** (Lenkwane La Mamphiri Orphanage): Organised recreational activities and provided mentorship to children, fostering a supportive environment outside of academic work.
- **Motivational Speaking:** Delivered talks at local schools to encourage students to pursue academic excellence despite challenges.